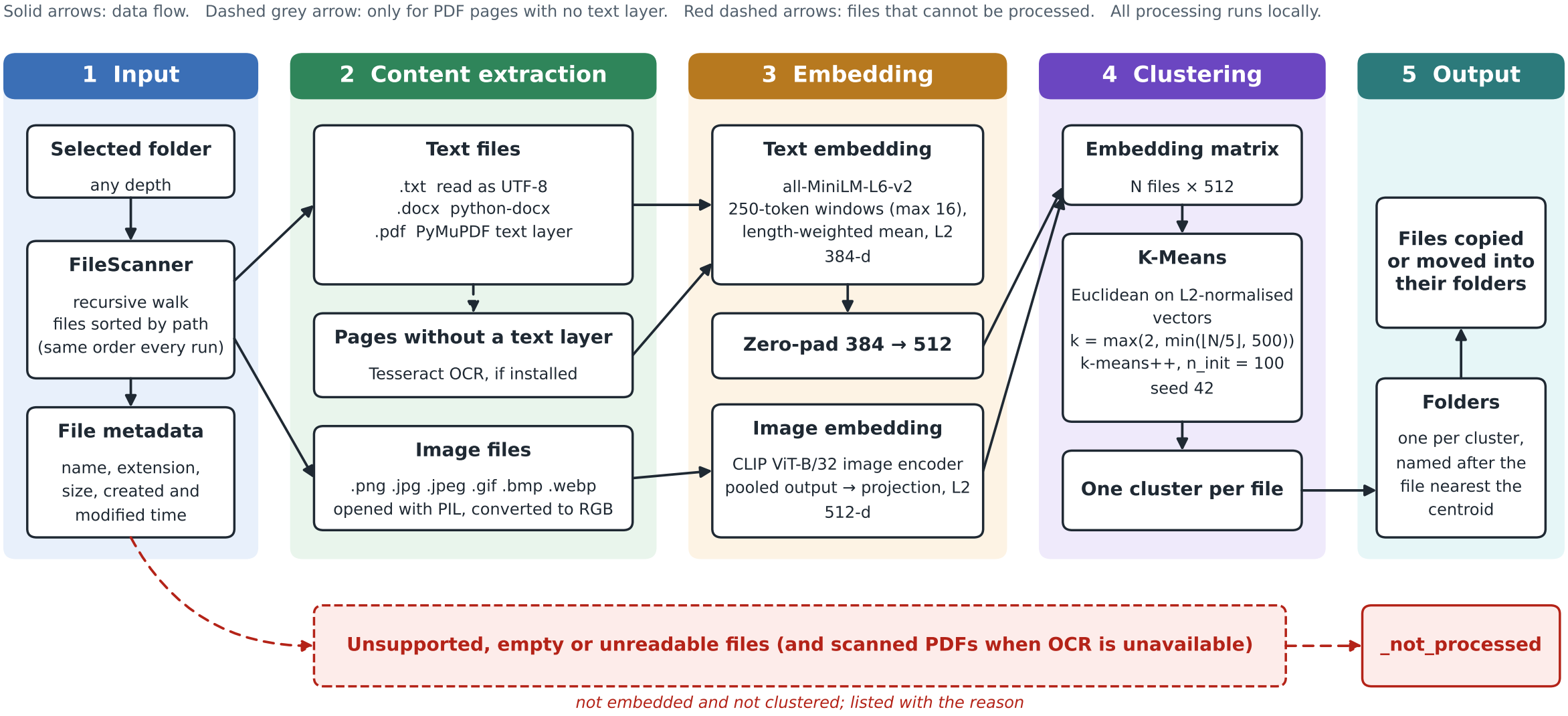
Solid arrows: data flow. Dashed grey arrow: only for PDF pages with no text layer. Red dashed arrows: files that cannot be processed. All processing runs locally.
1 Input
Selected folder
any depth
FileScanner
recursive walk
files sorted by path
(same order every run)
File metadata
name, extension,
size, created and
modified time
2 Content extraction
Text files
.txt read as UTF-8
.docx python-docx
.pdf PyMuPDF text layer
Pages without a text layer
Tesseract OCR, if installed
Image files
.png .jpg .jpeg .gif .bmp .webp
opened with PIL, converted to RGB
3 Embedding
Text embedding
all-MiniLM-L6-v2
250-token windows (max 16),
length-weighted mean, L2
384-d
Zero-pad 384 → 512
Image embedding
CLIP ViT-B/32 image encoder
pooled output → projection, L2
512-d
4 Clustering
Embedding matrix
N files × 512
K-Means
Euclidean on L2-normalised
vectors
k = max(2, min(⌊N/5⌋, 500))
k-means++, n_init = 100
seed 42
One cluster per file
5 Output
Files copied
or moved into
their folders
Folders
one per cluster,
named after the
file nearest the
centroid
Unsupported, empty or unreadable files (and scanned PDFs when OCR is unavailable)
_not_processed
not embedded and not clustered; listed with the reason